Customer Shopping Behaviour Analysis

1.Introduction

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The objective of the analysis is to understand customer demographics, spending patterns, product preferences, and purchasing behavior. These insights can help businesses make data-driven decisions related to marketing strategies, inventory planning, and customer engagement.

2.Dataset Summary

A summary of the dataset was generated using **df.info()** to understand the structure, data types, and statistical distribution of variables.

Dataset Size

- **Total Records:** 3,900 transactions
- **Total Columns:** 18 variables

Data Types

The dataset includes both **numerical and categorical variables**.

Numerical Columns

- Age
- Purchase Amount (USD)
- Review Rating
- Previous Purchases

Categorical Columns

- Gender
- Item Purchased
- Category
- Location
- Size
- Color
- Season
- Subscription Status
- Shipping Type
- Discount Applied
- Promo Code Used
- Payment Method
- Frequency of Purchases

Missing Values

The dataset contained **37 missing values**, all located in the **Review Rating** column. No missing values were found in other variables.

3.Exploratory Data Analysis using Python

Imported the dataset using pandas.

	customer_id	age	gender	item_purchased	category	purchase_amount	location	size	color	season	review_rating	subscription_status	shipping_type
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping
5	6	46	Male	Sneakers	Footwear	20	Wyoming	M	White	Summer	2.9	Yes	Standard
6	7	63	Male	Shirt	Clothing	85	Montana	M	Gray	Fall	3.2	Yes	Free Shipping
7	8	27	Male	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes	Free Shipping
8	9	26	Male	Coat	Outerwear	97	West Virginia	L	Silver	Summer	2.6	Yes	Express
9	10	57	Male	Handbag	Accessories	31	Missouri	M	Pink	Spring	4.8	Yes	2-Day Shipping

Before performing analysis, several data cleaning and preprocessing steps were applied to ensure data quality and consistency.

Handling Missing Values

The dataset contained **37 missing values** in the **Review Rating** column.
These missing values were handled using **median imputation within each product category**, ensuring ratings remain consistent with category trends.

```
#filling null values in rating column by grouping reviews by category and filling null values with their respective category
df["Review Rating"] = df.groupby("Category")["Review Rating"].transform(lambda x: x.fillna(x.median()))
```

Standardizing Column Names

To improve readability and maintain coding consistency, all column names were converted to snake_case format.

Creating Age Groups

To better analyze customer demographics, a new feature called **age_group** was created using quartile-based segmentation.

```
labels = ["Young Adult", "Adult", "Middle Aged", "Senior"]
df["age_group"] = pd.qcut(df["age"], q=4, labels = labels)
```

Customers were divided into four groups- Young Adult, Adult, Middle Aged, Senior.

Converting Purchase Frequency into Days

The **frequency_of_purchases** column contained categorical values such as *Weekly, Monthly, Quarterly,* etc.
To make the data more suitable for analysis, these values were converted into numeric equivalents representing the number of days between purchases.

```
frequency_mapping = { 'Fortnightly':14,
 'Weekly': 7,
 'Annually': 365,
 'Quarterly':90,
 'Bi-Weekly':14,
       'Monthly':30,
 'Every 3 Months': 90
}

df["purchase_frequency_days"] = df['frequency_of_purchases'].map(frequency_mapping)
```

Data Consistency Check:

Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

Database Integration

Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4.Data Analysis using SQL

After performing exploratory data analysis and preprocessing in Python, further analysis was conducted using **MySQL** to extract business insights from the customer shopping dataset.

1) Top 5 most purchased items.

item_purchased	Item_count
Blouse	171
Pants	171
Jewelry	171
Shirt	169
Dress	166

2) List top 5 locations with the highest number of customers.

location	Customers_count
Rhode Island	63
Kansas	63
Arizona	65
Hawaii	65
New Jersey	67

3) What is the total revenue generated by male vs. female customers?

gender	Revenue
Male	157890
Female	75191

4) Compare the average Purchase Amounts between Standard and Express Shipping.

shipping_type	round(avg(purchase_amount),2)
Express	60.48
Standard	58.46

5) Which 5 products have the highest percentage of purchases with discounts applied?

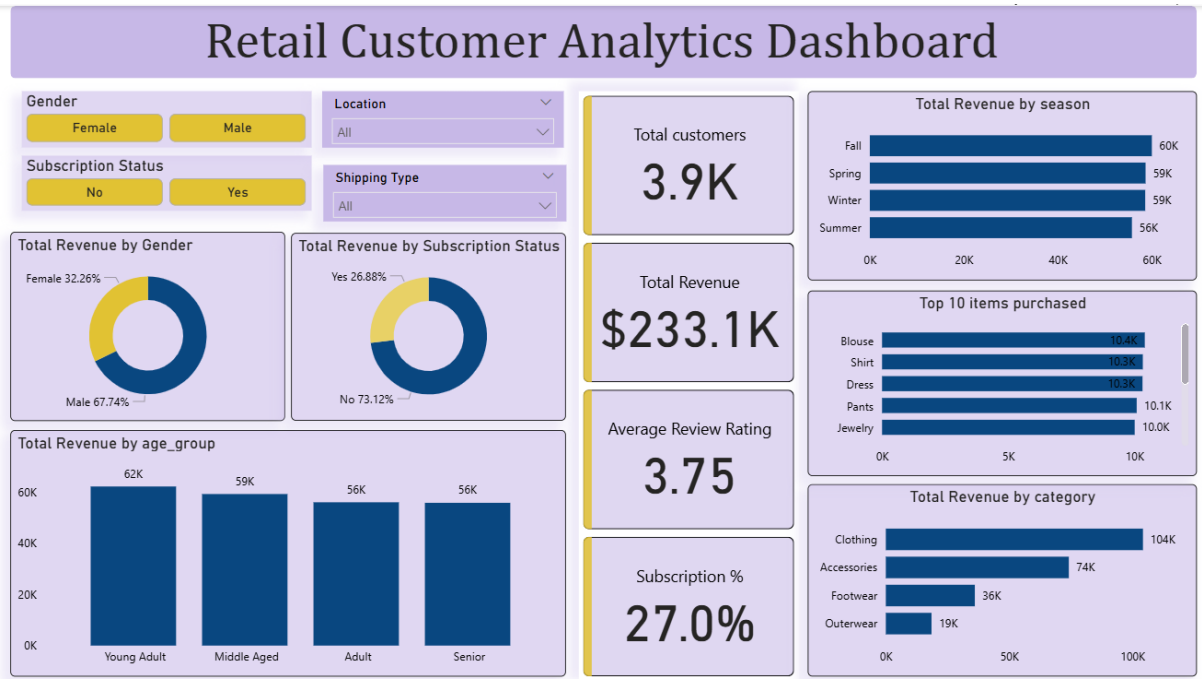
item_purchased	Discount_rate
Hat	50.0000
Sneakers	49.6552
Coat	49.0683
Sweater	48.1707
Pants	47.3684

6) Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

subscription_status	count(customer_id)
Yes	958
No	2518

5.Power BI Dashboard

To visualize the insights derived from the dataset, an interactive Power BI dashboard was created. The dashboard presents key metrics related to sales performance, customer behaviour, and product trends, allowing stakeholders to quickly explore patterns in customer purchases.



6.Business Recommendations

1)Strengthen Customer Retention Strategies

Customer segmentation analysis identified **New, Returning, and Loyal customers** based on previous purchase behaviour.

2) Expand the Subscription Program

The comparison between **subscribed and non-subscribed customers** suggests that subscription programs can drive higher engagement and consistent purchasing behavior.

3) Improve Discount Effectiveness

Discount analysis revealed that promotions influence customer spending but should be managed carefully to avoid reducing profit margins.

4)Promote High-Performing Products

The analysis identified **best-selling products and top-rated items** across different categories.

5) Focus Marketing on High-Spending Customer Segments

Analysis of spending patterns across **age groups and demographics** can help identify the most valuable customer segments.